Topological signatures from subsamples Central tendency properties of persistent homology can also be used to compute topological signatures for very large data sets, as an alternative approach to overcome the prohibitive cost of persistence computations. Given a large point cloud, the idea is to extract many subsamples, to compute the persistence landscape for each subsample and then to combine the information.

For any positive integer m, let $X = \{x_1, \cdots, x_m\}$ be a sample of m points drawn from a measure μ in a metric space M and which support is denoted $\mathbb{X}_\mu$. We assume that the diameter of $\mathbb{X}_\mu$ is finite and upper bounded by $\frac{T}{2}$, where T is the same constant as in the definition of persistence landscapes in Section 5.4. For ease of exposition, we focus on the case $k = 1$, and set $\lambda(t) = \lambda(1, t)$. However, the results we present in this section hold for $k > 1$. The corresponding persistence landscape (associated to the persistence diagram of the Čech or Rips-Vietoris filtration) is λ_X and we denote by Ψ_μ^m the measure induced by $\mu^{\otimes m}$ on the space of persistence landscapes. Note that the persistence landscape λ_X can be seen as a single draw from the measure Ψ_μ^m. The point-wise expectations of the (random) persistence landscape under this measure is defined by $\mathbb{E}_{\Psi_\mu^m}[\lambda_X(t)], t \in [0, T]$. The average landscape $\mathbb{E}_{\Psi_\mu^m}[\lambda_X]$ has a natural empirical counterpart, which can be used as its unbiased estimator. Let $S_1^m, \ldots, S_\ell^m$ be ℓ independent samples of size m from $\mu^{\otimes m}$. We define the empirical average landscape as

$$\overline{\lambda_\ell^m}(t) = \frac{1}{b} \sum_{i=1}^{b} \lambda_{S_i^m}(t), \qquad \text{for all } t \in [0, T], \tag{8}$$

and propose to use $\overline{\lambda_\ell^m}$ to estimate $\lambda_{\mathbb{X}_\mu}$. Note that computing the persistent homology of $\mathbb{X}_n$ is $O(\exp(n))$, whereas computing the average landscape is $O(b \exp(m))$.

Another motivation for this subsampling approach is that it can be also applied when μ is a discrete measure with support $\mathbb{X}_N = \{x_1, \ldots, x_N\}$ lying in a metric space M. This framework can be very common in practice, when a continuous (but unknown measure) is approximated by a discrete uniform measure μ_N on $\mathbb{X}_N$.

The average landscape $\mathbb{E}_{\Psi_\mu^m}[\lambda_X]$ is an interesting quantity on its own, since it carries some stable topological information about the underlying measure μ, from which the data are generated.

Theorem 11. [Chazal et al. (2015a)] *Let $X \sim \mu^{\otimes m}$ and $Y \sim \nu^{\otimes m}$, where μ and ν are two probability measures on M. For any $p \geq 1$ we have*

$$\left\| \mathbb{E}_{\Psi_\mu^m}[\lambda_X] - \mathbb{E}_{\Psi_\nu^m}[\lambda_Y] \right\|_\infty \leq 2\, m^{\frac{1}{p}} W_p(\mu, \nu),$$

where W_p is the pth Wasserstein distance on M.

The result of Theorem 11 is useful for two reasons. First, it tells us that for a fixed m, the expected "topological behavior" of a set of m points carries some stable information about the underlying measure from which the data are generated. Second, it provides a lower bound for the Wasserstein distance between two measures, based on the topological signature of samples of m points.

6.3.2 Asymptotic normality

As in the previous section, we consider several persistence diagrams (or other representations). The next step after giving central tendency descriptors of persistence homology is to provide asymptotic normality results for these quantities together with bootstrap procedures to derive confidence regions. It is of course easier to show such results for functional representations of persistence. In Chazal et al. (2013a, 2015b), following this strategy confidence bands for landscapes are proposed from the observation of landscapes $\lambda_1, \ldots, \lambda_N$ drawn i.i.d. from a random distribution in the space of landscapes. The asymptotic validity as well as the uniform

convergence of the multiplier bootstrap is shown in this framework. Note that similar results can be also proposed for many representations of persistence, in particular by showing that the corresponding functional spaces are Donsker spaces.

6.4 Other statistical approaches to TDA

Statistical approaches for TDA are knowing an increasing interest and many others have been proposed in the recent years or are still subject to active research activities, as illustrated in the following non-exhaustive list of examples.

Hypothesis testing. Several methods have been proposed for hypothesis testing procedures for persistent homology, mostly based on permutation strategies and for two sample testing. Robinson and Turner (2017) focuses on pairwise distances of persistence diagrams whereas Berry et al. (2020) study more general functional summaries. Hypothesis tests based on kernel approaches have been proposed in Kusano (2019). A two-stage hypothesis test of filtering and testing for persistent images is also presented inMoon and Lazar (2020).

Persistence Homology Transform. The representations introduced before are all transformations derived from the persistence diagram computed from a fixed filtration built over a data set. The Persistence Homology Transform introduced in Curry et al. (2018); Turner et al. (2014b) to study shapes in $\mathbb{R}^d$, takes a different path by looking at the persistence homology of the sublevel set filtration induced by the projection of the considered shape to each direction in $\mathbb{R}^d$. It comes with several interesting properties, in particular the Persistence Homology Transform is a sufficient statistic for distributions defined on the set of geometric and finite simplicial complexes embedded in $\mathbb{R}^d$.

Bayesian statistics for TDA. A Bayesian approach to persistence diagram inference has been proposed in Maroulas et al. (2020) by viewing a persistence diagram as a sample from a point process. This Bayesian method computes the point process posterior intensity based on a Gaussian mixture intensity for the prior.

6.5 Persistent homology and machine learning

Using TDA and more specifically persistent homology for machine learning is a subject that attracts a lot of information and generated an intense research activity. Although the recent progress in this area goes far beyond the scope of this paper, we briefly introduce the main research directions with a few reference to help the newcomer to the field to get started.

TDA **for exploratory data analysis and descriptive statistics.** In some domains, TDA can be fruitfully used as a tool for exploratory analysis and visualization. For example, the Mapper algorithm provides a powerful approach to explore and vizualize the global topological structure of complex data sets. In some cases, persistence diagrams obtained from data can be directly interpreted and exploited for better understanding of the phenomena from which the data have been generated. This is, for example, the case in the study of force fields in granular media (Kramar et al., 2013) or of atomic structures in glass (Nakamura et al., 2015) in material science, in the study of the evolution of convection patterns in fluid dynamics (Kramár et al., 2016), in machining monitoring (Khasawneh and Munch, 2016) or in the analysis of nanoporous structures in chemistry (Lee et al., 2017) where topological features can be rather clearly related to specific geometric structures and patterns in the considered data.

Persistent homology for feature engineering. There are many other cases where persistence features cannot be easily or directly interpreted but present valuable information for further processing. However, the highly non linear nature of diagrams prevents them to be immediately used as standard features in machine learning algorithms.

Persistence landscapes and linear representations of persistence diagrams offer a first option to convert persistence diagrams into elements of a vector space that can be directly used as features in classical machine learning pipelines. This approach has been used, for example, for protein binding (Kovacev-Nikolic et al., 2016), object recognition (Li et al., 2014) or time-series analysis. In the same vein, the construction of kernels for persistence diagrams that preserve their stability properties has recently attracted some attention. Most of them have been obtained by considering diagrams as discrete measures in $\mathbb{R}^2$. Convolving a symmetrized (with respect to the diagonal) version of persistence diagrams with a 2D Gaussian distribution, Reininghaus et al. (2015) introduce a multi-scale kernel and apply it to shape classification and texture recognition problems. Considering Wasserstein distance between projections of persistence diagrams on lines, Carriere and Oudot (2017) build another kernel and test its performance on several benchmarks. Other kernels, still obtained by considering persistence diagrams as measures, have also been proposed in Kusano et al. (2017).

Various other vector summaries of persistence diagrams have been proposed and then used as features for different problems. For example, basic summaries are considered in Bonis et al. (2016) and combined with quantization and pooling methods to address non rigid shape analysis problems; Betti curves extracted from persistence diagrams are used with 1-dimensional Convolutional Neural Networks (CNN) to analyze time dependent data and recognize human activities from inertial sensors in Dindin et al. (2020); Umeda (2017); *persistence images* are introduced in Adams et al. (2017) and are considered to address some inverse problems using linear machine learning models in Obayashi and Hiraoka (2017).

The above mentioned kernels and vector summaries of persistence diagrams are built independently of the considered data analysis or learning task. Moreover, it appears that in many cases the relevant topological information is not carried by the whole persistence diagrams but is concentrated in some localized regions that may not be obvious to identify. This usually makes the choice of a relevant kernel or vector summary very difficult for the user. To overcome this issue, various authors have proposed learning approaches that allows to learn the relevant topological features for a given task. In this direction, Hofer et al. (2017) proposes a deep learning approach to learn parameters of persistence images representations of persistence diagrams while Kim et al. (2020) introduce a neural network layer for persistence landscapes. In Carrière et al. (2020), the authors introduce a general neural network layer for persistence diagrams that can be either used to learn an appropriate vectorization or directly integrated in a deep neural network architecture. Other methods, inspired from k-means, propose unsupervised methods to vectorize persistence diagrams Royer et al. (2021); Zieliński et al. (2010), some of them coming with theoretical guarantees Chazal et al. (2020).

Persistent homology for machine learning architecture optimization and model selection. More recently, TDA has found new developments in machine learning where persistent homology is no longer used for feature engineering but as a tool to design, improve or select models - see, e.g. Carlsson and Gabrielsson (2020); Chen et al. (2019); Gabrielsson and Carlsson (2019); Hofer et al. (2019a); Moor et al. (2020); Ramamurthy et al. (2019); Rieck et al. (2019). Many of these tools rely on the introduction of loss or regularization functions depending on persistent homology features, raising the problem of their optimization. Building on the powerful tools provided by software libraries such as PyTorch or TensorFlow, practical methods allowing to encode and optimize a large family of persistence-based functions have been proposed and experimented Brüel-Gabrielsson et al. (2019); Poulenard et al. (2018). A general framework for persistence-based function optimization based on stochastic subgradient descent algorithms with